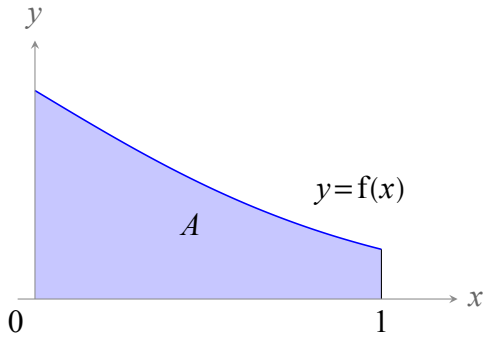
Compiled by Maximus Pung Jun
H2 Mathematics (9758)
2026

2026 H2 Math Definite Integrals Test: Solutions with comments

Total marks: 58

1 The diagram shows a sketch of the curve $y=\mathrm{f}(x)$. The shaded region under the curve between $x=0$ and $x=1$, as shown in the diagram, is A. This region is split into n vertical strips of equal width, h.



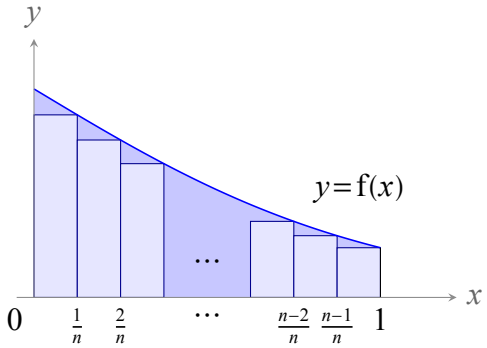
(a) State the value of h. [1]

(b) Hence show, using a sketch, that $\sum_{r=1}^{n} h\mathrm{f}(rh)$ is less than the area of A. [2]

You are now given that $\mathrm{f}(x)=\dfrac{\mathrm{e}^{-2x}}{1+\mathrm{e}^{-2x}}$.

(c) Hence find the exact value of $\lim_{n\to\infty}\sum_{r=1}^{n} h\mathrm{f}(rh)$. [3]

[ACJC/2026 Prelim/P1]

Solution		Comments
(a) **[1]**	$h=\frac{1}{n}$	
(b) **[2]**	 Total area of the n rectangles $=\frac{1}{n}\mathrm{f}\left(\frac{1}{n}\right)+\frac{1}{n}\mathrm{f}\left(\frac{2}{n}\right)+\frac{1}{n}\mathrm{f}\left(\frac{3}{n}\right)+\cdots+\frac{1}{n}\mathrm{f}\left(\frac{n}{n}\right)$ $=\sum_{r=1}^{n} h\mathrm{f}(rh)<A$ since the n rectangles (width $\frac{1}{n}$, right-end heights) lie under the curve.	
(c) **[3]**	As $n\to\infty$, the total area of the rectangles tends to the area of A. $\lim_{n\to\infty}\sum_{r=1}^{n} h\mathrm{f}(rh)=\int_0^1 \frac{\mathrm{e}^{-2x}}{\mathrm{e}^{-2x}+1}\,\mathrm{d}x$ $=\left[-\frac{1}{2}\ln\left\lvert\mathrm{e}^{-2x}+1\right\rvert\right]_0^1$ $=\frac{1}{2}\left[\ln(2)-\ln\left(1+\mathrm{e}^{-2}\right)\right]$	

2 A curve C has parametric equations

$$x=-\sqrt{t^2+4},\quad y=\frac{\ln t}{t},\quad \text{where } t>0.$$

(a) Show that $\dfrac{\mathrm{d}y}{\mathrm{d}x}=\dfrac{(\ln t-1)\sqrt{t^2+4}}{t^3}$. [3]

(b) The area of the region under a parametric curve from $t=\alpha$ to $t=\beta$ is given by the formula

$$\int_{\alpha}^{\beta} y\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)\mathrm{d}t.$$

Show that the area bounded by C and the lines $x=-\sqrt{13}$ and $x=-\sqrt{5}$ is given by

$$\int_{1}^{3}\frac{\ln t}{\sqrt{t^2+4}}\,\mathrm{d}t.$$

Find the area, giving your answer to 4 decimal places. [3]

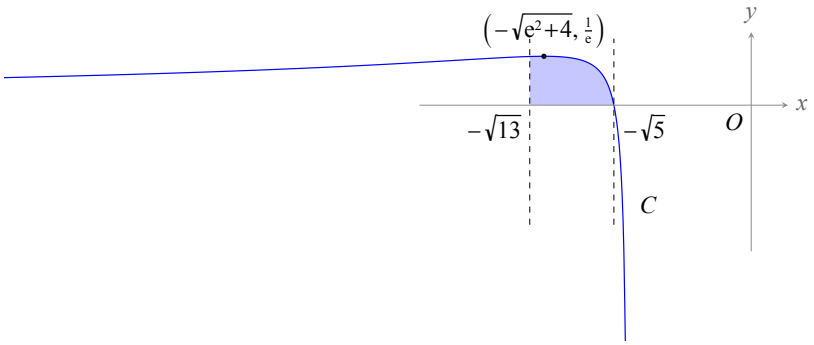
(c) It is given that the length of the arc of a curve defined parametrically by $x=\mathrm{f}(t), y=\mathrm{g}(t)$, from $t=t_1$ to $t=t_2$, is $\displaystyle\int_{t_1}^{t_2}\sqrt{\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)^2+\left(\frac{\mathrm{d}y}{\mathrm{d}t}\right)^2}\,\mathrm{d}t$.

Find the length of the arc of C from $t=1$ to $t=3$, giving your answer to 3 significant figures. [3]

[TMJC/2019 Prelim/P2 (Modified)]

Solution		**Comments**
(a) **[3]**	$x=-\sqrt{t^2+4} \Rightarrow \dfrac{\mathrm{d}x}{\mathrm{d}t}=-\dfrac{t}{\sqrt{t^2+4}}$ $y=\dfrac{\ln t}{t} \Rightarrow \dfrac{\mathrm{d}y}{\mathrm{d}t}=\dfrac{t\left(\frac{1}{t}\right)-\ln t}{t^2}=\dfrac{1-\ln t}{t^2}$ $\dfrac{\mathrm{d}y}{\mathrm{d}x}=\dfrac{\mathrm{d}y/\mathrm{d}t}{\mathrm{d}x/\mathrm{d}t}=\dfrac{1-\ln t}{t^2}\times\left(-\dfrac{\sqrt{t^2+4}}{t}\right)$ $=\dfrac{(\ln t-1)\sqrt{t^2+4}}{t^3}$ (shown)	

(b)
[3]



When $x=-\sqrt{13}$: $t^2+4=13 \Rightarrow t=3$ (since $t>0$).

When $x=-\sqrt{5}$: $t^2+4=5 \Rightarrow t=1$.

For $1 \le t \le 3$, $\ln t \ge 0$, so $y \ge 0$ and the region is above the x-axis.

$$\text{Area} = \int_{-\sqrt{13}}^{-\sqrt{5}} y \, \mathrm{d}x$$

$$= \int_3^1 \frac{\ln t}{t}\left(-\frac{t}{\sqrt{t^2+4}}\right) \mathrm{d}t$$

$$= \int_1^3 \frac{\ln t}{\sqrt{t^2+4}} \, \mathrm{d}t \quad \text{(shown)}$$

$$= 0.431675$$

$$= 0.4317 \text{ units}^2 \quad (4 \text{ d.p.})$$

(c)
[3]

$$\text{Arc length} = \int_1^3 \sqrt{\left(-\frac{t}{\sqrt{t^2+4}}\right)^2 + \left(\frac{1-\ln t}{t^2}\right)^2} \, \mathrm{d}t$$

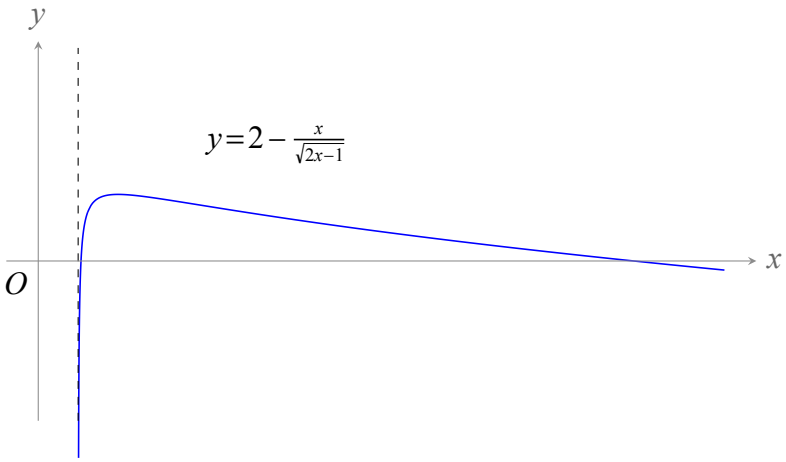
$$= \int_1^3 \sqrt{\frac{t^2}{t^2+4} + \frac{(1-\ln t)^2}{t^4}} \, \mathrm{d}t$$

$$= 1.503666$$

$$= 1.50 \text{ units} \quad (3 \text{ s.f.})$$

3 **(a)** Using the substitution $u=\sqrt{2x-1}$, show that $\int \frac{x}{\sqrt{2x-1}}\,\mathrm{d}x=\int \frac{u^2+1}{2}\,\mathrm{d}u$. [2]

(b) **(i)** The diagram below shows the graph of $y=2-\frac{x}{\sqrt{2x-1}}$. Show, on a copy of this diagram, the equation of the asymptote of the curve and the coordinates of its turning point and points of intersection with the x-axis. [2]



(ii) The region R is bounded by the curve $y=2-\frac{x}{\sqrt{2x-1}}$ and the lines $x=1$, $x=3$ and $y=\frac{1}{2}$. Find the exact area of R. [5]

(c) The region S is bounded by the curves $y=2-\frac{x}{\sqrt{2x-1}}$, $x=2(y-1)^2+1$, the lines $x=1$, $x=4$ and the x-axis.

(i) Express y in terms of x for the curve $x=2(y-1)^2+1$, giving both branches. [1]

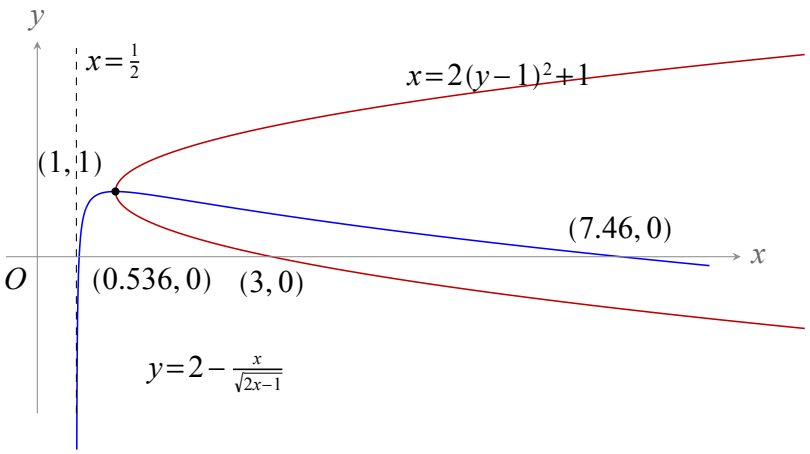
(ii) On the same diagram as in part **(b)(i)**, sketch the graph of $x=2(y-1)^2+1$. [2]

(iii) Find the volume of the solid generated when S is rotated through 2π radians about the x-axis. Give your answer correct to 3 decimal places. [3]

[RI/2025 Prelim/P1]

Solution		Comments
(a) **[2]**	$u=\sqrt{2x-1} \Rightarrow u^2=2x-1$ Hence $u\dfrac{\mathrm{d}u}{\mathrm{d}x}=1 \Rightarrow \dfrac{\mathrm{d}x}{\mathrm{d}u}=u.$ $\displaystyle\int \frac{x}{\sqrt{2x-1}}\,\mathrm{d}x=\int \frac{\frac{u^2+1}{2}}{u}\left(\frac{\mathrm{d}x}{\mathrm{d}u}\right)\mathrm{d}u$ $\displaystyle=\int \frac{u^2+1}{2u}(u)\,\mathrm{d}u=\int \frac{u^2+1}{2}\,\mathrm{d}u$ (shown) Alternatively, $u=\sqrt{2x-1} \Rightarrow \dfrac{\mathrm{d}u}{\mathrm{d}x}=\dfrac{1}{\sqrt{2x-1}}$ and therefore $\displaystyle\int \frac{x}{\sqrt{2x-1}}\,\mathrm{d}x=\int x\cdot\frac{1}{\sqrt{2x-1}}\,\mathrm{d}x$ $\displaystyle=\int \frac{u^2+1}{2}\left(\frac{\mathrm{d}u}{\mathrm{d}x}\right)\mathrm{d}x=\int \frac{u^2+1}{2}\,\mathrm{d}u$	

(b)(i) [2]



Note: Detailed workings for intercepts and turning points (Optional, since more efficient to use GC)

When $y=0$,

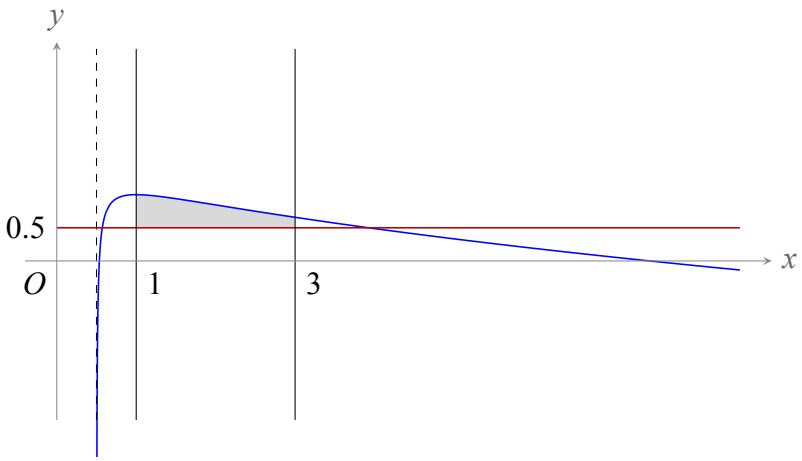
$$\frac{x}{\sqrt{2x-1}}=2$$

$$\Rightarrow x^2=4(2x-1)$$

$$\Rightarrow x=4\pm 2\sqrt{3}$$

$$\frac{dy}{dx}=-\frac{\sqrt{2x-1}-\frac{x}{\sqrt{2x-1}}}{2x-1}=\frac{1-x}{\left(\sqrt{2x-1}\right)^3}=0\Rightarrow x=1\Rightarrow y=1$$

(b)(ii)
[5]



Required area of R

$$= \int_1^3 \left(2 - \frac{x}{\sqrt{2x-1}} - \frac{1}{2}\right) \mathrm{d}x$$

$$= \int_1^3 \frac{3}{2}\,\mathrm{d}x - \int_1^3 \frac{x}{\sqrt{2x-1}}\,\mathrm{d}x$$

$u = \sqrt{2x-1}$
When $x = 3$, $u = \sqrt{5}$
When $x = 1$, $u = \sqrt{1} = 1$

$$= \frac{3}{2}[x]_1^3 - \int_1^{\sqrt{5}} \frac{u^2+1}{2}\,\mathrm{d}u$$

$$= 3 - \frac{1}{2}\left[\frac{u^3}{3} + u\right]_1^{\sqrt{5}}$$

$$= 3 - \frac{1}{2}\left[\frac{5\sqrt{5}}{3} + \sqrt{5} - \frac{1}{3} - 1\right]$$

$$= 3 - \left[\frac{4\sqrt{5}}{3} - \frac{2}{3}\right]$$

$$= \frac{11 - 4\sqrt{5}}{3}$$

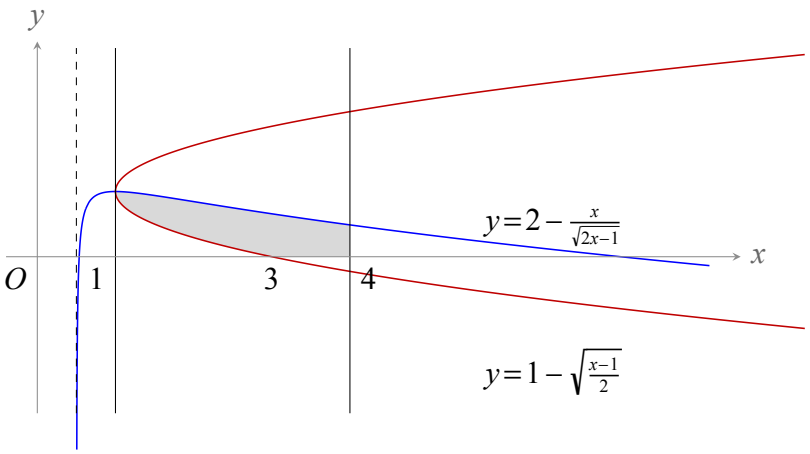
(c)(i)
[1]

$$x = 2(y-1)^2 + 1 \Rightarrow y = 1 \pm \sqrt{\frac{x-1}{2}}$$

(c)(ii)
[2]

Please refer to part **(b)(i)**.

(c)(iii)
[3]



Required volume

$$=\pi\int_1^4\left(2-\frac{x}{\sqrt{2x-1}}\right)^2\,\mathrm{d}x-\pi\int_1^3\left(1-\sqrt{\frac{x-1}{2}}\right)^2\,\mathrm{d}x$$

$=4.5183329$

$=4.518$ units^3 (3 d.p.)

4 In this question $\sin^{-1}(1-x)$ refers to the principal value.

(a) **(i)** Use the substitution $z=\sin^{-1}(1-x)$ to show that $\int_0^1 \sin^{-1}(1-x)\,\mathrm{d}x=\int_a^b z\cos z\,\mathrm{d}z$, where a and b are constants to be determined. [2]

(ii) Hence show that $\int_0^1 \sin^{-1}(1-x)\,\mathrm{d}x=\frac{\pi}{2}-1$. [2]

(b) A curve C is given by the equation

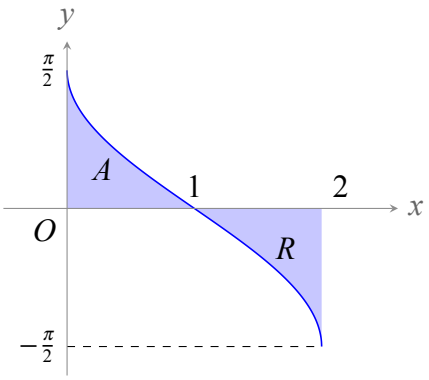
$$y=\sin^{-1}(1-x) \quad \text{for } 0\le x\le 2.$$

(i) Find exactly $\int_0^2 \left|\sin^{-1}(1-x)\right|\,\mathrm{d}x$. [2]

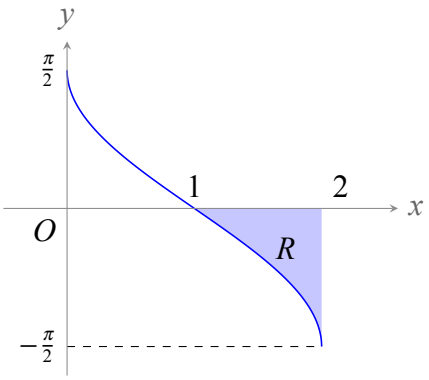
(ii) The region R is bounded by the curve C, the line $x=2$ and the part of the x-axis between 1 and 2. Find the exact volume generated when R is rotated 360° about the y-axis. [4]

[ACJC/2026 Prelim/P2]

Solution		Comments
(a)(i) **[2]**	$z=\sin^{-1}(1-x)$ $\sin z=1-x$ $\cos z\,\frac{\mathrm{d}z}{\mathrm{d}x}=-1$ When $x=0$, $z=\sin^{-1}1=\frac{\pi}{2}$. When $x=1$, $z=\sin^{-1}0=0$. $\int_0^1 \sin^{-1}(1-x)\,\mathrm{d}x=\int_{\frac{\pi}{2}}^0 -z\cos z\,\mathrm{d}z$ $=\int_0^{\frac{\pi}{2}} z\cos z\,\mathrm{d}z$ $a=0$ and $b=\frac{\pi}{2}$.	
(a)(ii) **[2]**	By parts, with $u=z$, $\frac{\mathrm{d}v}{\mathrm{d}z}=\cos z$: $\int_0^{\frac{\pi}{2}} z\cos z\,\mathrm{d}z=[z\sin z]_0^{\frac{\pi}{2}}-\int_0^{\frac{\pi}{2}}\sin z\,\mathrm{d}z$ $=\frac{\pi}{2}+[\cos z]_0^{\frac{\pi}{2}}$ $=\frac{\pi}{2}-1$ (shown)	

(b)(i) **[2]**	 Since the curve $y=\sin^{-1}(1-x)$ is symmetrical about $(1,0)$, area A = area R. $\int_0^2 \left\lvert \sin^{-1}(1-x) \right\rvert \, dx = 2\int_0^1 \sin^{-1}(1-x)\, dx$ $= 2\left(\frac{\pi}{2}-1\right) = \pi - 2$	

(b)(ii) [4]



$y=\sin^{-1}(1-x) \Rightarrow \sin y=1-x \Rightarrow x=1-\sin y$

When $x=2, y=\sin^{-1}(-1)=-\frac{\pi}{2}$. When $x=1, y=\sin^{-1}0=0$.

$$\text{Volume}=\pi(2)^2\left(\frac{\pi}{2}\right)-\pi\int_{-\frac{\pi}{2}}^{0}x^2\,\mathrm{d}y$$

$$=2\pi^2-\pi\int_{-\frac{\pi}{2}}^{0}(1-\sin y)^2\,\mathrm{d}y$$

$$=2\pi^2-\pi\int_{-\frac{\pi}{2}}^{0}\left(1+\sin^2 y-2\sin y\right)\mathrm{d}y$$

$$=2\pi^2-\pi\int_{-\frac{\pi}{2}}^{0}\left(\frac{3}{2}-\frac{1}{2}\cos 2y-2\sin y\right)\mathrm{d}y$$

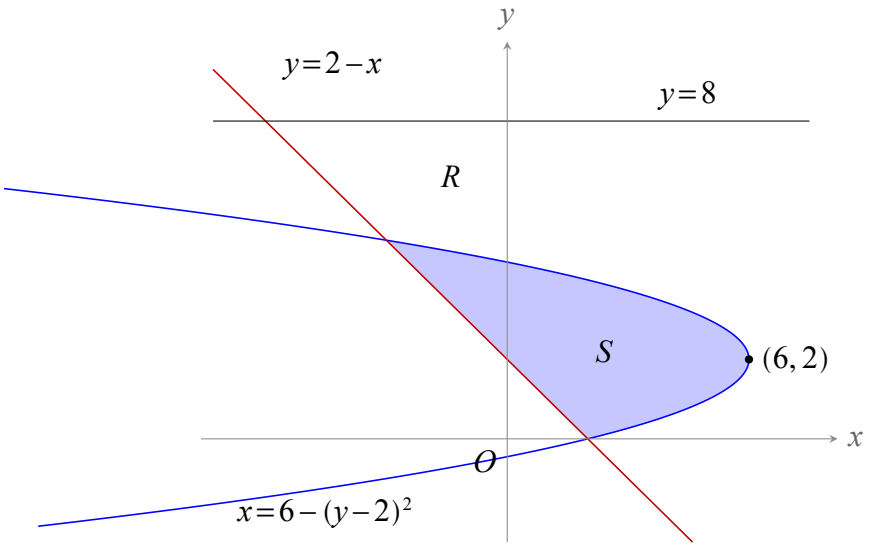
$$=2\pi^2-\pi\left[\frac{3y}{2}-\frac{1}{4}\sin 2y+2\cos y\right]_{-\frac{\pi}{2}}^{0}$$

$$=2\pi^2-\pi\left[2-\left(-\frac{3\pi}{4}\right)\right]$$

$$=2\pi^2-2\pi-\frac{3\pi^2}{4}$$

$$=\frac{5\pi^2}{4}-2\pi \text{ units}^3$$

5 In the diagram below, the region R is bounded by the curve C with equation $x=6-(y-2)^2$, the lines $y=8$, $y=2-x$ and the y-axis. The region S is bounded by C and the line $y=2-x$.



(a) Find the exact area of region R. [5]

(b) Find the volume of the solid of revolution formed when region S is rotated through 360° about the x-axis, leaving your answer to 2 decimal places. [3]

[VJC/2025 Prelim/P1]

Solution		**Comments**
(a) **[5]**	For point of intersection of C and $y=2-x$: $2-y=6-(y-2)^2$ $2-y=6-y^2+4y-4$ $y^2-5y=0$ $y(y-5)=0$ $y=0$ or $y=5$ When $x=0$, $6-(y-2)^2=0 \Rightarrow y=2\pm\sqrt{6}$. Area of R	

$$= -\left[\int_{2+\sqrt{6}}^{5} (6-(y-2)^2)\,\mathrm{d}y + \int_{5}^{8} (2-y)\,\mathrm{d}y\right]$$

OR $\left(\frac{6+3}{2}\right)\times 3 - \int_{2+\sqrt{6}}^{5} (6-(y-2)^2)\,\mathrm{d}y$

$$= -\left\{\left[6y - \frac{(y-2)^3}{3}\right]_{2+\sqrt{6}}^{5} + \left[2y - \frac{y^2}{2}\right]_{5}^{8}\right\}$$

$$= -\left[(30-9) - \left(6\left(2+\sqrt{6}\right) - \frac{\left(\sqrt{6}\right)^3}{3}\right) + \left(16-\frac{64}{2}\right) - \left(10-\frac{25}{2}\right)\right]$$

$$= \frac{9}{2} + 4\sqrt{6} \text{ units}^2$$

Alternative method:

For point of intersection of C and $y=2-x$:

$$2-y = 6-(y-2)^2$$
$$2-y = 6-y^2+4y-4$$
$$y^2-5y=0$$
$$y(y-5)=0$$
$$y=0 \text{ or } y=5 \therefore x=2 \text{ or } -3$$

When $y=8$, $8=2-x \Rightarrow x=-6$.

Area of R

$$= (6)(8) - \int_{-6}^{-3} (2-x)\,\mathrm{d}x - \int_{-3}^{0} \left(2+\sqrt{6-x}\right)\mathrm{d}x$$

$$= 48 - \left[2x - \frac{x^2}{2}\right]_{-6}^{-3} - \left[2x - \frac{(6-x)^{\frac{3}{2}}}{\frac{3}{2}}\right]_{-3}^{0}$$

$$= 48 - \frac{39}{2} - \left(\left(-\frac{2}{3}\right)(6)^{\frac{3}{2}} - \left(-6-\frac{2}{3}(9)^{\frac{3}{2}}\right)\right)$$

$$= \frac{57}{2} - \left(24 - \frac{2}{3}\left(6\sqrt{6}\right)\right)$$

$$= \frac{9}{2} + 4\sqrt{6}$$

(b)
[3]

$$x = 6-(y-2)^2$$
$$y = 2 \pm \sqrt{6-x}$$

$$\text{Volume of solid} = \pi\int_{-3}^{6} \left(2+\sqrt{6-x}\right)^2 \mathrm{d}x - \pi\int_{-3}^{2} (2-x)^2\,\mathrm{d}x - \pi\int_{2}^{6} \left(2-\sqrt{6-x}\right)^2 \mathrm{d}x$$

$$= 327.25 \text{ units}^3$$

6 A curve C has equation $y = xe^{-x}$, $x \geq 0$.

(a) Show that C has exactly one stationary point. [2]

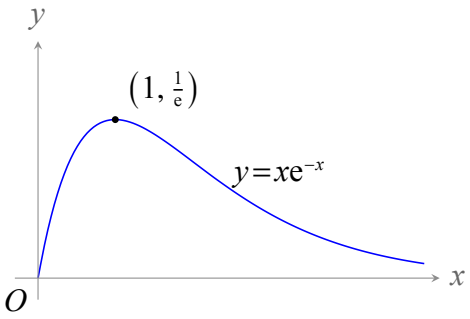
(b) Sketch C, giving the exact coordinates of the stationary point. [1]

The region R is bounded by C, the y-axis and the line $y = \frac{1}{e}$.

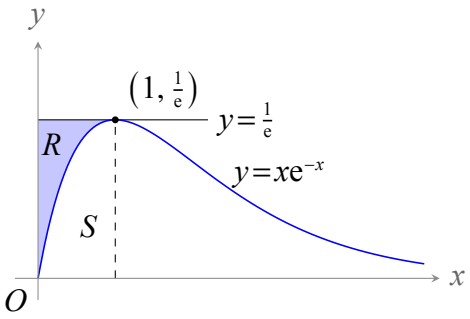
(c) Find the exact area of R. [4]

(d) An ornament is formed by rotating R through 2π radians about the line $y = 2$. By considering a suitable translation of C, find the volume of the ornament. [3]

[RI/2026 Prelim/P1]

Solution		Comments
(a) **[2]**	$y = xe^{-x}$ $\frac{dy}{dx} = e^{-x} + x(-e^{-x}) = e^{-x}(1-x)$ When $\frac{dy}{dx} = 0$, $e^{-x}(1-x) = 0$. Since $e^{-x} > 0$ for all $x \in \mathbb{R}$, $1 - x = 0 \Rightarrow x = 1$. Hence, C has exactly one stationary point. (shown)	
(b) **[1]**	 When $x = 1$, $y = \frac{1}{e}$; stationary point $\left(1, \frac{1}{e}\right)$; $y \to 0$ as $x \to \infty$.	

(c)
[4]



By parts, with $u=x$, $\frac{dv}{dx}=e^{-x}$:

$$\text{Area of } S=\int_0^1 xe^{-x}\,dx$$

$$=[-xe^{-x}]_0^1-\int_0^1 -e^{-x}\,dx$$

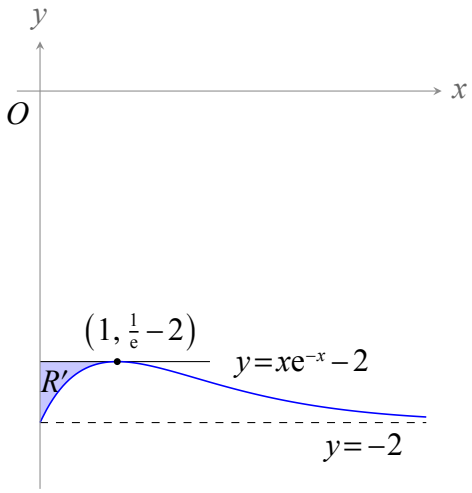
$$=-e^{-1}+[-e^{-x}]_0^1$$

$$=-\frac{1}{e}+\left(-\frac{1}{e}+1\right)=1-\frac{2}{e}$$

$$\text{Area of } R=\text{Area of rectangle}-\text{Area of } S$$

$$=(1)\left(\frac{1}{e}\right)-\left(1-\frac{2}{e}\right)$$

$$=\frac{3}{e}-1 \text{ units}^2$$

(d) **[3]**	Translate C 2 units in the negative y-direction: $y=xe^{-x}-2$.  Volume of ornament = Volume generated when R is rotated through 2π radians about the line $y=2$ = Volume generated when R' is rotated through 2π radians about the line $y=0$ $=\pi\int_0^1\left[(xe^{-x}-2)^2-\left(\frac{1}{e}-2\right)^2\right]dx$ $=1.13 \text{ units}^3$ (3 s.f.) **OR** $=\pi\int_0^1(xe^{-x}-2)^2\,dx-\pi\left(2-\frac{1}{e}\right)^2$ (1) $=1.13 \text{ units}^3$ (3 s.f.)	

END OF SOLUTIONS